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Basketball shooting angle calculation and analysis by deeply-learned vision model

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ABSTRACT

To solve the problem that standard image decomposition cannot conduct image decomposition of basketball shooting angle according to the change of basketball projection Angle, an intelligent visual image decomposition method of basketball projection angle using odd-difference value decomposition was proposed. By analyzing the change of basketball shooting angle under different shooting conditions, the basketball shooting angle is determined. Based on this, singular value decomposition is introduced to judge the image decomposition of intelligent vision. Observations have shown that the intelligent vision image decomposition judgment method can improve the image contrast of shooting motion decomposition. In the shortest time, according to the change of basketball projection angle, the shooting motion image decomposition is fulfilled. In order to improve the accuracy of human motion recognition in videos and computation of large-scale data sets, a motion recognition technique by leveraging double-core extreme learning machine and deep learning technology are proposed. In the first layer of the double-core extreme learning component, the linear kernel extreme learning machine is used to predict the dense trajectory characteristics and deep learning characteristics. In the second layer, dense trajectory features are combined with deep learning features. In deep features, deep video features are fused with video RGB trichromatic features as deep features. Simulation experiments are conducted on a massive-scale and real-world data sets and a small basketball gesture data set. The experimental results have shown that our method exhibits a high recognition accuracy for both data sets. The analysis results have shown that the improved model has higher recognition rate and better action recognition effect than its competitors.

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1. Introduction

In the motion of basketball shooting, different shooting angles will be produced owing to the diverse human shooting movements. The shot image captured by the traditional image decomposition method cannot characterize the clear image of every moment of angle change in the process of shooting angle change. This may causes the phenomenon of poor contrast and long time spent in the decomposition process of images processed by the existing intelligent vision decomposition technique. During basketball shooting, the shooting angle in the shooting of basketball can be divided into the angle when throwing the hollow basket and the angle when hitting the backboard [1]. Different shooting styles produce different angles, and different determination methods are required when handling different angles. Since the shooting angle is determined during basketball projection, the singular value is decomposed according to the determined angle. Meanwhile, the image decomposition judgment is described according to the decomposition results. Thus, the shooting angle vision-based decomposition and judgment has becoming a useful technique. Experiments have shown that the model proposed by us can significantly improve existing methods. The traditional methods explicitly divide video action recognition into three components: feature extraction, feature aggregation and feature classification. First, the underlying features of the video are extracted by the proposed algorithm, and subsequently the extracted features are encoded. Finally, the encoding results are categorized by classifiers such as SVM [2] and Adaboost [3], followed by PCA [4]. The commonly used characteristics of the bottom layer include SIFT [5], SIFT-3D [6], HOG [4], HOG3D [7], HOF [8], MBH [9], DT [10], IDT [5], etc. SIFT (Scale Invariant Feature Transform) calculates the directions and sizes of feature points extracted on various scale spaces as video descriptions. HOG (Histogram of Oriented Gradients) takes the histogram of the gradient direction of the statistical local image as the video feature.

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SIFT-3D and HOG3D extended the time dimension on the basis of SIFT and HOG respectively. HOF (Histogram of Optical Flow) calculates the optical basis of each video frame, and further describes the video through the histogram of optical flow information that statistics the included the angle between the optical flow vector and the X axis. MBH (Motion Boundary Histogram) combines HOG and HOF to calculate HOG features using the optical flow diagrams of X axis and Y axis, which exhibits better robustness than the previous two. DT (Dense Trajectory) collects feature points in single frames at various scales through grid partition. It abandons the inefficient feature points and subsequently extracts the above features from these points. Currently, the most effective manual feature is IDT (improved Dense Trajectory) based on DT. Compared with DT, IDT method improves the feature regularization, optimizes the optical flow, and facilitates the feature coding. Pervasively used methods for feature aggregation include BOVW [9], FV [10], Vlad [11], etc. BOVW (bag-of-visual Words) applies K-means to cluster features and replaces feature points by the nearest cluster center. This leads to a large loss of information. FV (Fisher Vector) uses the GMM method to model the feature points, and utilizes the linear combination of all clustering centers as the representation of the feature points. It has a high accuracy and an intolerable computation. Also, some information will be lost in the modeling process. VLAD (Vector of Locally Aggregated Descriptor) calculates the residuals from the feature points to the cluster center. After summation and normalization of the residuals falling on the same cluster center, a VLAD vector is obtained accordingly (as shown in Fig. 1).

2. Related work

For human motion recognition based on machine learning, Kapathy et al. [2] used deep convolutional neural network to recognize human motion with continuous RGB visual frequency frames as the direct input. Shuwang Ji et al. [3] used the temporal-level dimension information in video data and 3D convolutional neural network to recognize human movements. Jeff et al. [4] proposed a network model for human action recognition using a long-term recurrent convolutional network (LRCN) that integrates the convolutional layer and the long-term recurrent convolutional layer. Human motion recognition using video data source is to divide videos into time frame images, and then classify these images that vary over time. Therefore, deep learning method in the field of image recognition is also applied in human motion recognition toward video sequences [7]. However, motion recognition has the attributes of unpredictability and diversity. Thus, it is necessary to combine the multi-aspect comprehensive techniques. By increasing the concurrent data in practice, it brings certain difficulty to human motion recognize. In order to improve human motion recognition, an improved model based on the convolutional neural network model and the recursive neural network model is constructed, based on which the optical flow of the frames before and after the video is extracted by the to recognize the human motion in the visual frequency under the specific field scene. With the explosive growth of computational storage and rapid development of network recently, videos have gradually became a kind of data that people face daily. Image recognition has made great progress since the well-known AlexNet in 2012. The excellent performance of deep learning in image recognition also drives the rapid development of video action recognition. Video motion recognition, as a relatively young and challenging problem in machine vision, requires the application of image processing and machine learning. Video motion recognition has excellent commercial value in the long run. It is used for intelligent surveillance in the Internet of Things. It is also utilized for human–computer interaction in three-dimensional games. In the aspect of intelligent retail, it is used for customer behavior analysis, etc. However, the acquired video data is usually not the data under ideal conditions. It might be disturbed by various background factors such as shadow, body shaking, and background movement. Therefore, the investigation of human motion recognition with complex backgrounds is also a very important challenge. Video action recognition, namely video classification, differs from image recognition in temporal domain implied by multiple video frames. In order to make use of temporal information. There are mainly a combination of dual-stream Network, 3D Neural Network, Recurrent Neural Network (RNN) [1] and Convolutional Neural Network (CNN) [2] Three kinds of algorithms. Based on the double-stream network, we employ the non-local method [4,7], which is similar to the self-attention mechanism [3] in machine translation. We propose the non-local time interval network and the improved hybrid non-local time interval network for video classification. On the basis of the two-stream network, our method can make the network emphasizes the wider range of information by adding non-local computation. The optical flow information can be incorporated into the non-local computation to make the network pay attention to the action region. This can enhance the robustness to the complex static background of video. In addition, in order to integrate the multiplex discrimination results of two-stream segmenting networks, the weighted combinatorial square method which is sensitive to all kinds of actions is proposed.

Deep learning for human action recognition Based has achieved a breakthrough in image-related tasks compared with traditional methods. Videos, as an extension of images at temporal-level, has been extensively processed and analyzed by deep learning methods recently. The biggest difference between video and image lies in that it contains the message of the fourth dimension — time dimension besides length, width and channel. An optimal video action recognition method is bound to fully leverage the temporal information. Deep learning has achieved a major breakthrough in image-related tasks compared with traditional methods. Video, as an extension of images in temporal domain, has been extensively analyzed by deep learning methods in recent years. The biggest difference between video and image lies in that it contains the message of the fourth dimension — time dimension besides length, width and channel. A good video action recognition method is bound to make full use of the information in the time domain. Two-stream Convnet [11] is pervasively used for action recognition. It is a simple but has quite excellent effect. Generally, the dual-stream network consists of two channels: time stream network inputs single or stacked frames of video into the convolutional network to learn the video information in the spatial domain. The time stream network inputs optical flow diagram into the convolutional network in the form of multiple channels to learn the video information at temporal-level. The probability of the video output of the two networks respectively belongs to various types. Finally, the final classification result is obtained by averaging the two probability vectors. The modified dual-flow network [12] explores how and when spatial and temporal flow networks merge. It is very helpful for the fusion effect to map the spatial feature map of a specific region to the corresponding region of the time feature map. Therefore, the fusion should be conducted at the early stage of the network, rather than only at the end of the fusion of the double-stream information. This will lose the useful information. There are other ways to use audio information as a separate network, but it has never become mainstream. The two-stream convolutional network is based on 2D network of image recognition. Therefore, it can be pre-trained on the ImageNet as image recognition to improve the training effect and reduce the training time.

Wang et al. [9] proposed a novel trajectory pooling network (TDD) by combining the manual characteristics of classical IDT

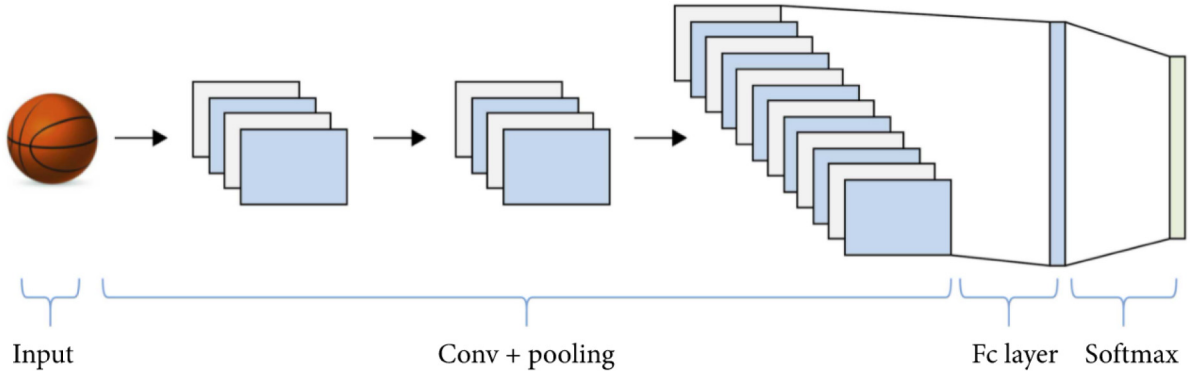

Fig. 1. Pipeline of our proposed basketball action recognition framework.

and the double-flow depth characteristics. TDD utilizes the two-stream networks to learn the convolution features. It applies the trajectory pooling to fuse these convolution features, and finally uses an SVM to classify them. Later, many methods are combined with IDT to achieve the best results. 3D neural network [8] is another commonly used categorization model. With the improvement of computational power and the increase of data set size, 3D neural network has been developed. The 3D neural network was inferior to the conventional dual-stream network. 3D network is designed as an end-to-end deep architecture that takes video directly as input and outputs the final categories. Due to the complex network structure, 3D network requires a large data set to achieve the satisfactory results. The number of network layers should not be very deep. Sun et al. [4] decomposed "3D space–time learning" into "2D spatial learning" and "1D time learning". Then they proposed a spatial–temporal decomposition convolutional network. Literature [7–9] has been improved on this basis. I3D [10] fills the parameters of the pre-trained 2D network into 3D and adopts a double-flow structure. Each path is a 3D network, unlike the traditional double-flow network, which is a 2D network. Combining the advantages of dual-stream network and 3D oracle network, i3D has achieved the best performance in many well-known data sets. RNN can process spatial–temporal information, so the combination of CNN and RNN can also learn the temporal-level signal in visual frequency. LSTM (Long Short-Term Memory) [5] and GRU (GRU) [6] are two varieties pervasively used in RNN. Its basic idea is to use RNN to integrate the last layer of CNN in the timeline. It does not fuse the last feature after the full connection layer of CNN, because the pooling of high-level features after the full connection layer has lost the information of spatial features on the timeline. At present, the Longterm Recurrent Convolutional neural Network (LRCN) can achieve the best result [12].

3. Our proposed method

The framework of the kernel extreme learning machine in this paper is divided into two layers, as shown in Fig. 1. In the first layer, feature cores of deep learning and manual feature cores are fused, and the output of the first layer is the predicted score of the three feature cores. The second layer trains the classifier, which maps all the predicted scores into the final action classification. Fusion of feature nuclei Fusion of different feature nuclei can contain features of different dimensions of video. Therefore, manual feature kernel and deep learning feature kernel are combined in this paper, and L2 universal number is used to calculate linear kernel. A linear kernel matrix can be defined as

$$K(5, z,) = \hat{}(z).\hat{}1 \quad (1)$$

where K(5, ∞) is the (I, j) th element of K. The fusion feature kernel is calculated by calculating the average value of the kernel matrix of different feature sources. After the fusion of feature kernel, three cores were obtained: deep learning feature kernel, manual feature kernel and fusion feature kernel. Then, the prediction scores of different characteristic nuclei are calculated by using kernel extreme learning machine. It is assumed that the predicted scores can be combined to calculate the final action classification of the video sequence. Therefore, the three output score vectors in this paper are fused into a score vector S. If {s:,☆ }, I 1 1…, row represents the training data set, where the number of behavioral training samples, s: ∈ Rm is the combined prediction score, q is the total number of action categories. It is a real classification of actions. Considering that the output of the first layer is taken as the input feature of the second layer, this paper uses L1 universal number to regularize the output of the first layer. After that, this paper obtains a feature vector of each video and calculates the kernel matrix at the second layer. Since the performance of radial basis function kernel for L1 norm features is better than that of linear kernel, radial basis function kernel (RBF) is adopted in this paper. The radial basis function kernel (RBF) is defined as the following equation K(S). Herein, s, and s are the predicted scores of video I and j respectively. Note that K(party s,) is the (I, J) element of RBF kernel K. Referring to the conclusion in [13], the performance of nonlinear kernel based on histogram features is better than that of linear kernel. For manual features, IDT (improved dense track) descriptor of video is extracted based on open source code. IDT descriptor includes track line, HOG (static feature description), H()F(pixel absolute motion feature), MBH (pixel relative motion feature). Then, Fisher vector is used to encode the IDT feature. Since Fisher feature requires encoding the first-order statistics and second-order statistics between the video descriptor and the Gaussian mixture model, the amount of data is large. This paper designs a mechanism to reduce the dimension of feature descriptor: The main component analysis is performed on the descriptor, and the number of Gaussian elements K is set as 256. A subset containing 256 OOO features is randomly sampled from the training set to train the GMM (Gaussian Mixture Model) model. Finally, each descriptor type for each video is represented as a 2×D×K dimensional Fisher vector, where D represents the descriptor dimension processed by principal component analysis. For example, the final descriptor dimension of the trackline is 2×15×256–7680. The L2 norm is used to process the Fisher vector, and the open software library [14] is used to calculate GMM and Fisher vector. Since the calculation efficiency of linear kernel of descriptor is higher, which is much higher than that of distance metric such as Distance of Distance, this paper does not connect descriptors into unified vectors, but calculates the linear kernel of each descriptor independently. Finally, the kernel matrix K 1 of manual features

is calculated as the mean value K^1 of the kernel matrix of descriptor K. For deep learning features, this paper designs deep learning features based on motion information and Convolutional Neural Network (Convolutional Neural Network). The scheme in reference [12] is used to organization deep learning features into a 4096-dimensional video descriptor, which is processed by L2 norm. Then, the linear kernel of deep learning features is calculated, and the kernel matrix K of deep learning features is established. In the process of nuclear fusion, The average value of deep learning feature core and manual feature quotient is calculated to obtain the result of nuclear fusion: K one (%‰+K^)/2. Finally, the nuclear extreme learning machine in this paper uses three features to check the input video stream for action classification processing, and the three feature cores are: Manual feature kernel, deep learning feature kernel and fusion kernel. The open source kernel extreme learning machine code base is used to realize the two-layer kernel extreme learning machine algorithm in this paper. After the kernel extreme learning machine parameters are obtained through grid search, the other two unknown parameters are set as: C1-1, D1-10 [11].

Time Selection LSTM Network RNN is the best architecture for learning short - and long-term action sequence dependencies, which can potentially capture time information.

$$t = \sigma(WXFXT + \ WHFHT - 1 + WCI\ CT - 1 + \ BI) \quad (2)$$

$$FT = \sigma(WXFXT + \ WHFHT - 1 + WCF\ CT - 1 + \ BF) \quad (3)$$

$$Ct = IT * TANH(WXCXT + Bi)WHCHT - 1 + BC, +FT * CT - 1 \quad (4)$$

$$OT = \sigma(Wxoxt + \ WHOHT - 1 + WCO\ ct + \ BO) \quad (5)$$

where f_t and o_t respectively correspond to the output of the input gate, the forgetting gate and the output door at time t; W and B are the weight matrix and bias vector of the corresponding door, respectively. O represents the multiplication of elements in the corresponding position of the vector; The symbol σ stands for the sigmoid activation function.

After the classification and recognition of the above three types of neural networks, a Dropout layer is added to eliminate the overfitting phenomenon. After the training of the LSTM model, the scale of characteristic information obtained is 1×64. A full link layer is added to connect all nodes in the current layer with all nodes in the previous layer completely. Then, the human actions in the results are classified by Softmax classifier.

4. Experimental results and analysis

In order to comprehensively evaluate the performance of this algorithm, experiments were carried out on UCFL01 video data set and NATOPS video data set respectively. UCFL01 data set has a low resolution and a large amount of data, which can test the recognition performance of the algorithm for large-scale data sets. The NATOPS video data set has a high resolution and contains a total of 24 movements. These movements are human upper limb movements with a small amplitude, and some movements overlap the upper limb and the body. This data set can test the recognition effect of this algorithm on small amplitude movements. 4.1 Experiment of UCFLOL Data set 4.1.1 Introduction of Data set UCFL01 Dataset was collected on the Internet. The complexity of the data set was high, and all the videos had obvious background clutter. UCFL01 contains 13 320 video clips with a total of 101 action categories. In this paper, the default partitioning scheme of 3 training sets and 1 test set is adopted for the data set. For each partitioning scheme, 7 video sequences are selected from 25 subgroups as test sequences, and the other 18 video sequences are selected as training sequences. Experiments

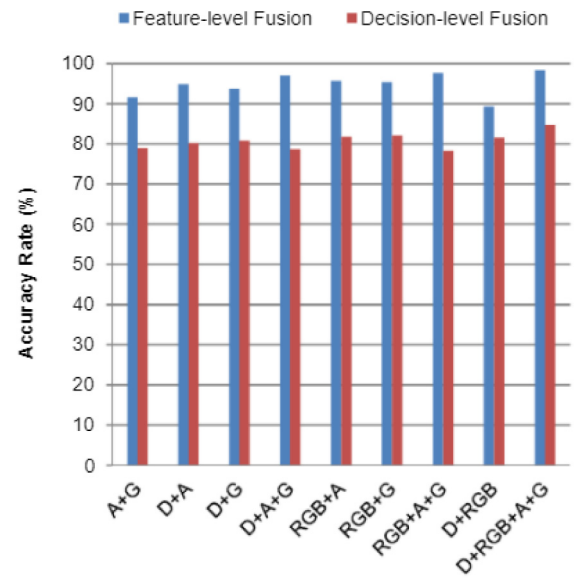

Fig. 2. Recognition accuracy of the baseline recognition algorithms.

Model	Modality	Accuracy
OF-RGB-CNN	RGB	40,6%
OF-RGB-CNN-SVM	RGB	**44%**
OF-Depth-CNN	Depth	46,9%
OF-Depth-CNN-SVM	Depth	**50,2%**
OF-RGB-Depth-CNN-SVM	RGB+Depth	**65%**

Fig. 3. Accuracy of the baseline recognition models.

were carried out on the three default partitioning schemes of UCFL01 data set, and the average value of the three groups of data was taken as the final experimental result.

In order to evaluate the recognition performance of this algorithm for UCFL01 data set, this algorithm is compared with other action recognition algorithms, which are as follows: Action recognition algorithm based on convolution neural network (CNN) [5], action recognition algorithm based on improved dense trajectory (*III* T) [10], action recognition algorithm based on depth of learning (C3D) [11] |, based on optical flow and convolution gesture recognition algorithm of neural network - T (CNN) [1 ", based on temporal motion information and gesture recognition algorithm of support vector machine (sVM) o | 5. Different algorithms of action recognition results as shown in Fig. 3, you can see CNN - T, the sVM and the recognition rate of this algorithm is better than that of CNN, IDT and C3D the three algorithms, CNN-T, SVM and this algorithm all belong to the recognition algorithm of multi-feature fusion, while CNN, IDT and C3D are all recognition algorithms of single feature, and it can be concluded that: Multiple feature fusion recognition performance is better than that of single feature recognition performance. In addition, the recognition rate of this algorithm is a little better than CNN - T, S, M two algorithms, the algorithm with the SVM algorithm is relatively similar, main area do not lies in the movement of the algorithm was designed based on event template information mechanism. The system can effectively improve the precision of gesture recognition (as shown in Fig. 2). (See Fig. 3)

Two-stream Convnet [12] is a very common method in the field of action recognition, which is simple but has quite excellent effect. As the name suggests, the dual-stream network consists

of two channels: time stream network inputs single or stacked frames of video into the convolutional network to learn the video information in the spatial domain, while time stream network inputs optical flow diagram into the convolutional network in the form of multiple channels to learn the video information in the temporal domain. The probability of the video output of the two networks respectively belongs to various types, and finally the final classification result is obtained by averaging the two probability vectors. The modified dual-flow network [21] explores how and when spatial and temporal flow networks merge. In his opinion, it is very helpful for the fusion effect to map the spatial feature map of a specific region to the corresponding region of the time feature map. Therefore, the fusion should be carried out at the early stage of the network, rather than only at the end of the fusion of the double-stream information, which will lose a lot of useful information. There are other ways to use audio information as a separate network, but it has never become mainstream. The two-stream convolutional network is based on 2D network of image recognition, so it can be pre-trained on ImageNet as image recognition to improve the training effect and reduce the training time. Wang et al. [22] proposed trajectory pooling network (TDD) by combining the manual characteristics of classical IDT and the double-flow depth characteristics. TDD uses two-stream networks to learn the convolution features, then uses trajectory pooling to fuse these convolution features, and finally uses SVM to classify them. Later, many methods are combined with IDT to achieve the best results.

5. Conclusions

In this paper, an improved network model for human movement recognition in video is proposed. In this model, the optical flow data of video images is extracted by using dense optical flow function. Then, the neural network model based on deep learning is improved. The model presented in this paper has better discrimination effect, and the recognition rate is superior to other models. In this paper, when comparing the recognition rate of each scene, it is found that the total recognition rate is different in different scenes, and the recognition rate is low in the context of the change of the distance and near scale in outdoor. The prediction is due to the fact that the change of the distance and near scale of the lens affects the continuity of information in feature extraction. In the future, further experimental research will be conducted on this problem.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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